

ModernTSF Real Data Visualization and Analysis

This report visualizes real time-series data in a ModernTSF-style data analysis workflow.

Raw CSV: /data/jm/tsorchestra_repro_20260610/finance1000_source/financial_hourly_1000_raw.csv

Log-return CSV: /data/jm/tsorchestra_repro_20260610/finance1000_source/financial_hourly_1000_logret.csv

Universe: 1000 tickers

Timestamps: 28510

Span: 2015-01-02 00:00:00+00:00 to 2026-02-25 00:00:00+00:00

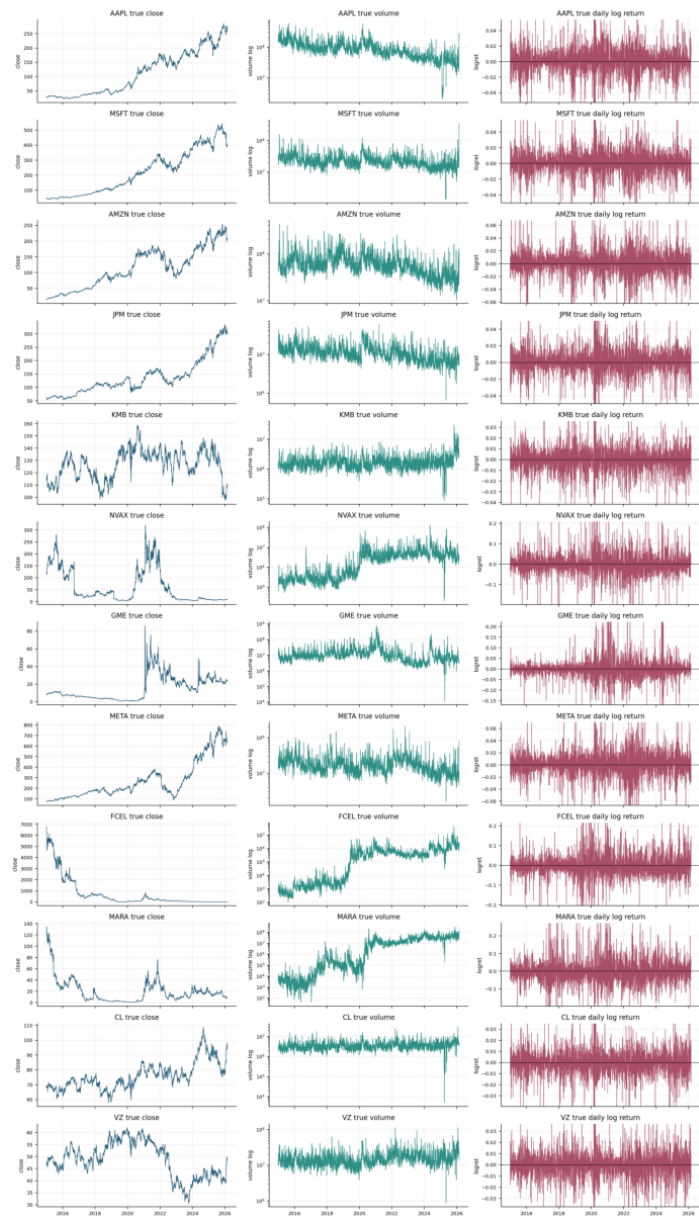
Representative tickers: AAPL, MSFT, AMZN, JPM, KMB, NVAX, GME, META, FCEL, MARA, CL, VZ

Included figures:

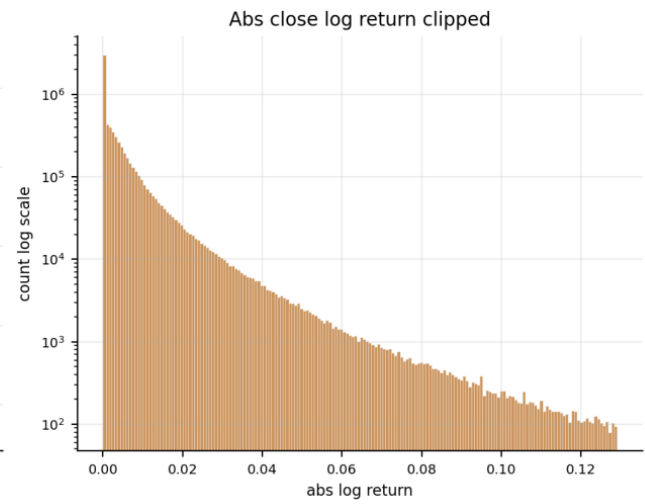
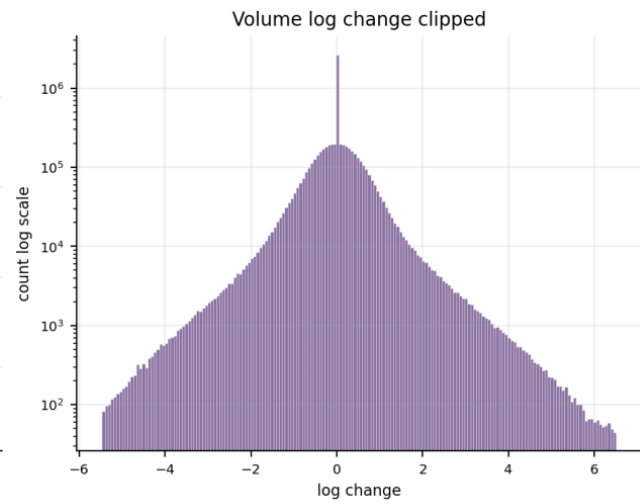
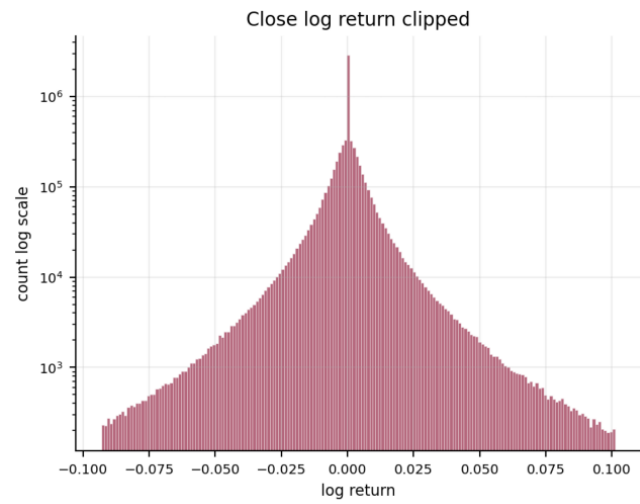
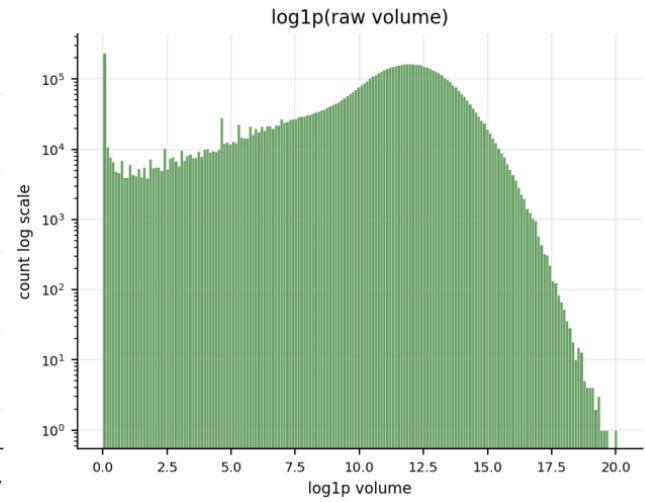
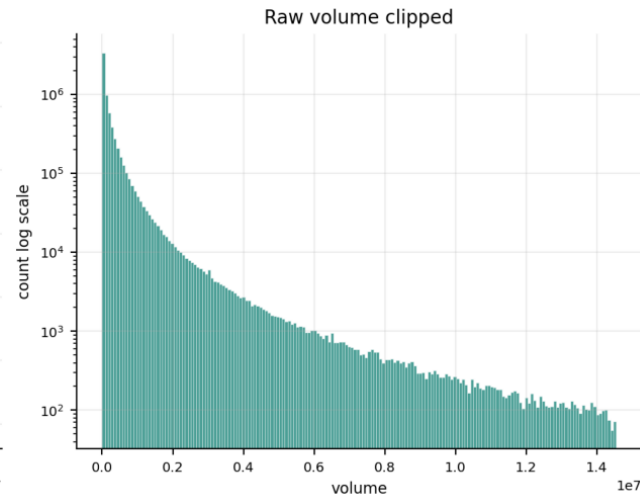
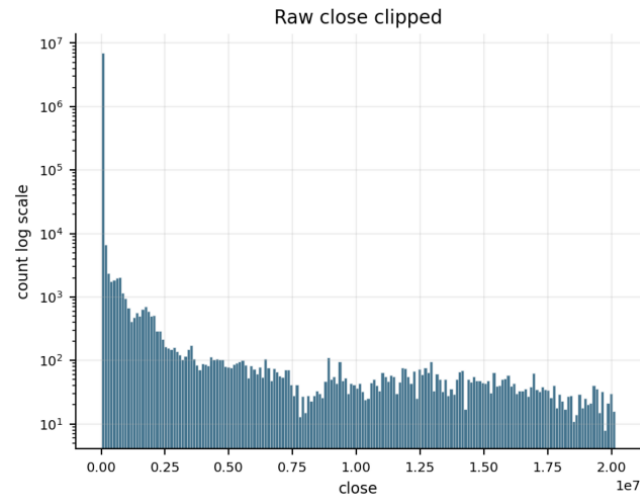
1. Real close, volume, and log-return curves by ticker
2. Raw and log-return distribution panels
3. Log-return QQ plot, tail CCDF, and tail event rates
4. Cross-sectional close and volume structure over time
5. Per-ticker volatility, volume, and return heterogeneity
6. Log-return and rolling-volatility heatmaps

Ticker book: ticker_book_all_1000.pdf

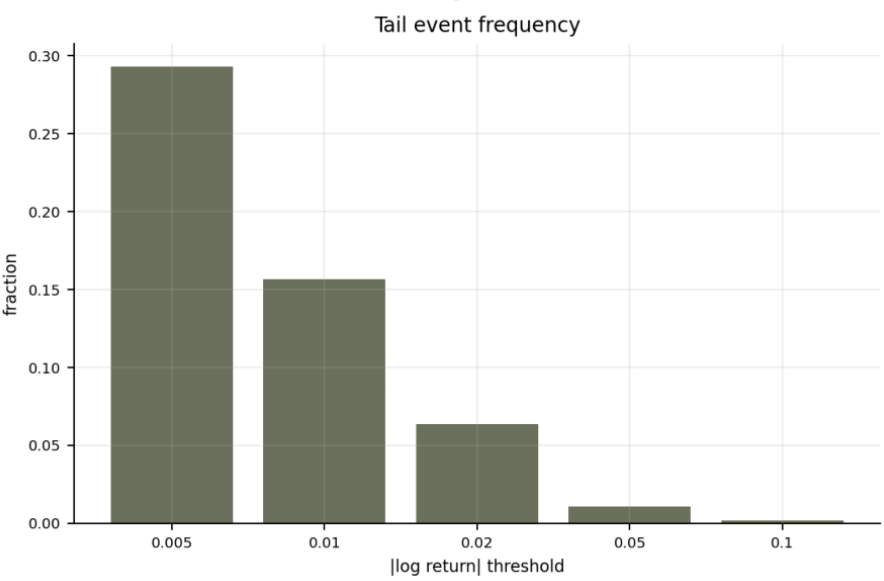
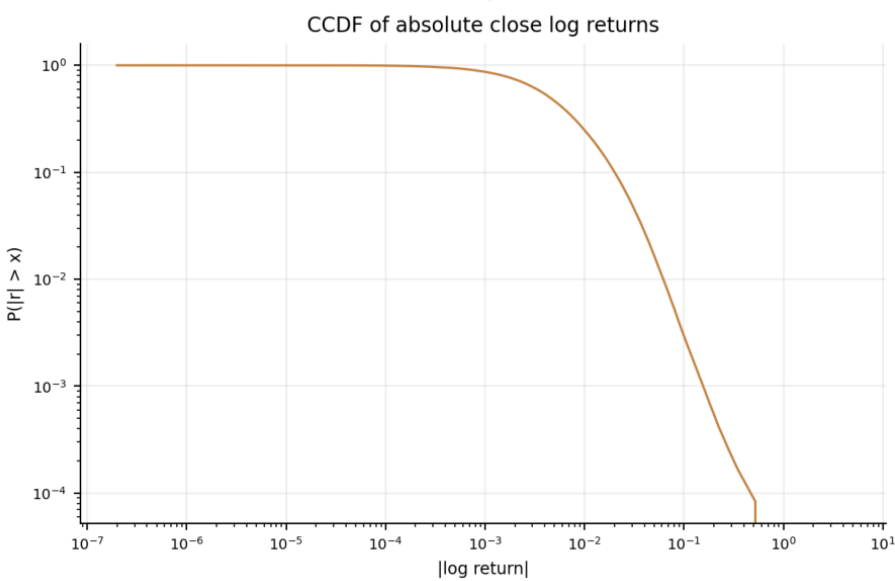
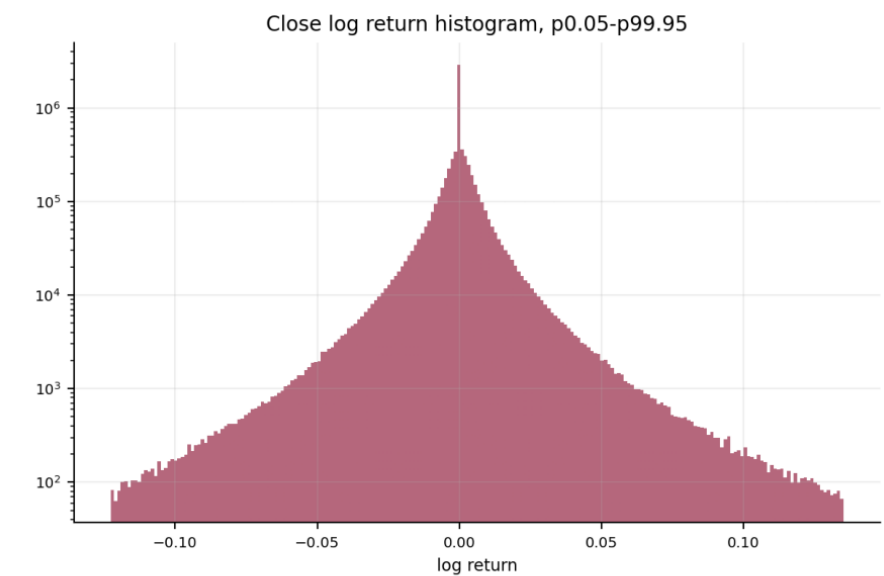
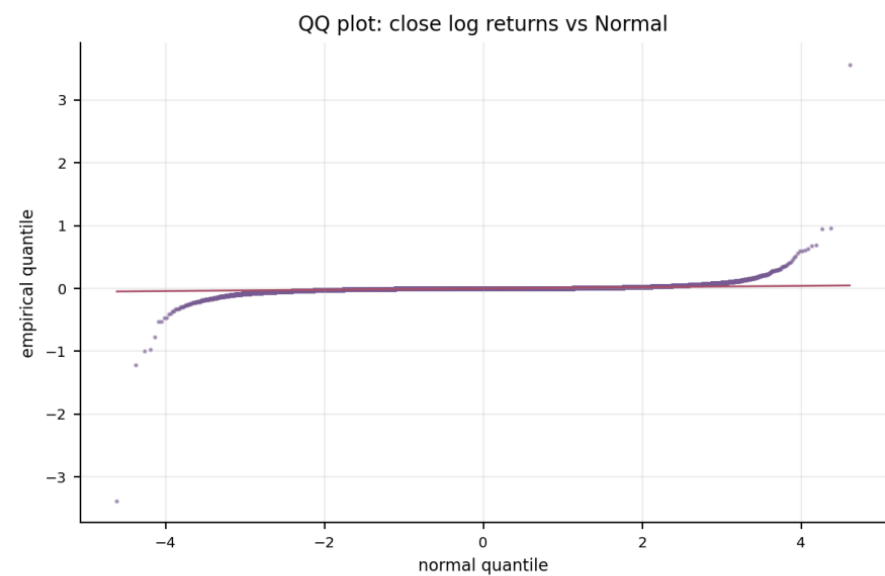
Real Historical Curves: Close, Volume, and Log Return



Empirical Distributions from Real Data



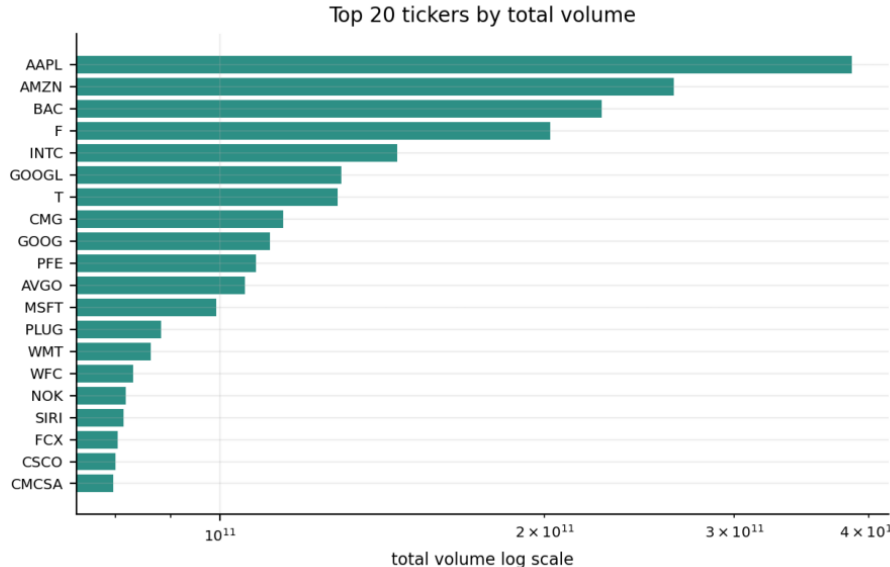
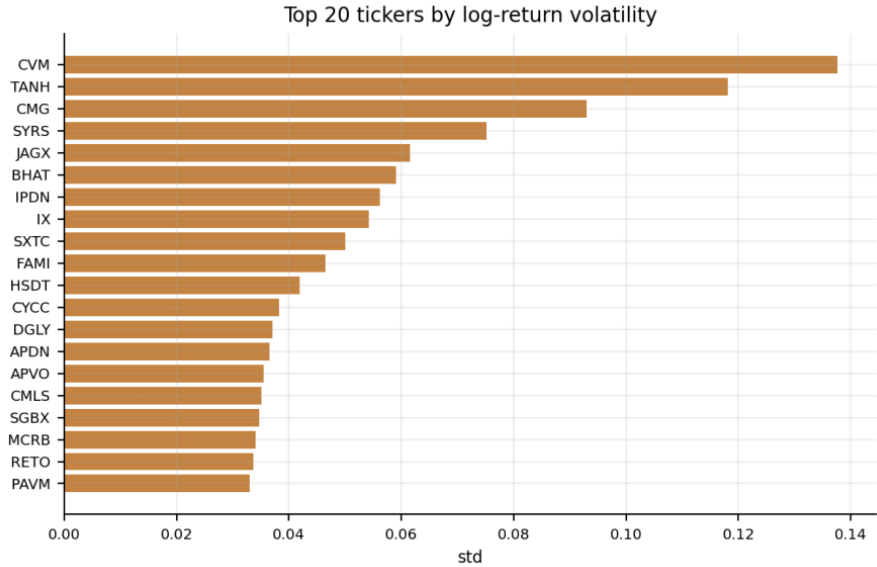
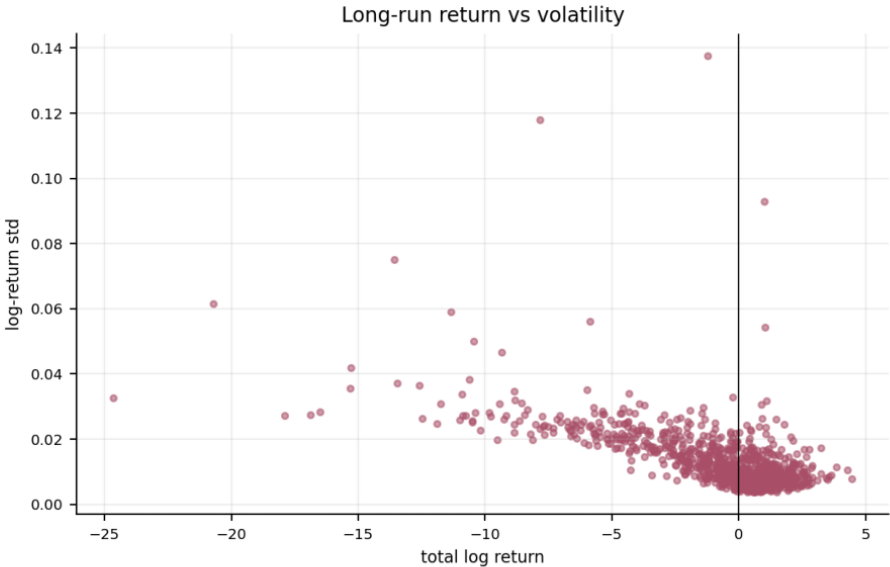
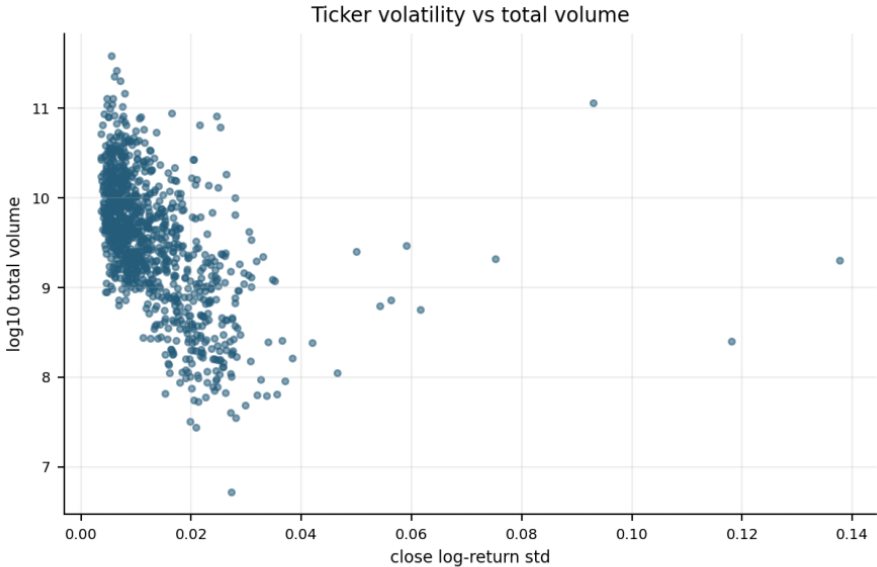
Log Return Diagnostics: Distribution, Normality, and Tails



Cross-Sectional Structure in Raw Data

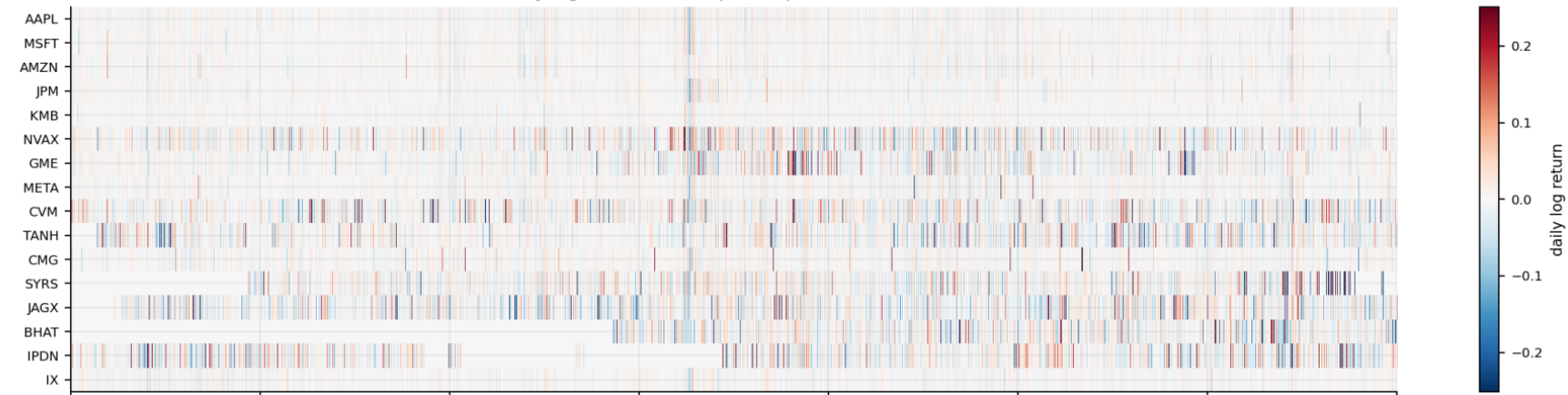


Per-Ticker Heterogeneity



Time Structure in Log Returns and Volatility

Daily log-return heatmap for representative tickers



30-day rolling annualized volatility heatmap

